Experimental Economics Coursework 2 (80%):

The Question:

Does reducing moral wiggle room through objective cross-validation decrease the strategic deflation of peer ratings in one-shot group assessments?

Introduction

The free-riding problem is common in group projects with shared rewards and individual effort. The problem of free riding is not limited to the simple example of classrooms; it's a general problem in public goods. The most common answer to this problem is peer assessment. The concept of peer assessment is to assign grades to team members to redistribute the total reward based on each member's participation and effort. This solution is based on the assumption that, in a group project, members can observe the amount of effort put in by the rest of the team. However, this system has a major weakness: it breaks down if team members lie, since an agent's rating of the other influences their grade, and the points are redistributed if there is no mechanism to prevent it. In peer assessment, the dominant strategy is to deflate all peers' ratings to artificially inflate the player's rating. This is allowed by the ambiguity of the rating since there is no system to check whether your peer assessment seems reasonable; agents have the "moral wiggle room" to convince themselves that all the members performed poorly. The first question when this is observed is when peer assessment works.

The most efficient way to avoid misreporting is to combine peer assessment with repeated collaboration among agents. The fact that the agent is going to collaborate multiple times makes inflating or deflating another peer's rating a justification for retaliation. This repetition of collaboration activates direct reciprocity in the threat of retaliation Kandel & Lazear, (1992). The issue is that repeated collaboration is uncommon in educational settings. To reduce misreporting during a one-time collaboration with peer assessment, one solution is to implement institutional mechanisms like cross-validation or a flagging system. These can identify if your rating of a peer significantly deviates from the overall consensus. It works by reducing moral wiggle room, thereby activating the self-image channel, according to Dana, Weber & Kuang (2007). Their research demonstrates that removing transparency enables players to make more selfish choices while maintaining the illusion of fairness.

Those two mechanisms theoretically yield the same result by reducing free-riding and incentivising truthful reporting, but it remains unanswered whether they suppress misreporting in the same way.

Since repeated interaction is rarely achieved in higher education contexts, cross-validated peer assessment seems the most appropriate solution. To provide strong evidence of the efficiency of the cross-validation system, an experimental economics test is necessary. This raises the question of whether the supposed effectiveness of cross-validated peer assessment is observed in a context of experimental economics.

This experiment aims to compare two conditions: a controlled condition, which is a one-shot peer assessment game, and the second condition. This is the classic model that certain universities use; we will use it as a baseline to assess the effectiveness of the second

condition. The second is the treatment condition with the exact same condition; the only difference will be the presence of the cross-validation mechanism.
Since the repetition condition is rarely fulfilled in the academic context, an alternative institutional design is required. This experiment aims to assess whether cross-validated peer assessment can effectively suppress the missreporting in a purely one-shot context.

Literature review:
Kandel & Lazear (1992) explore free-riding in a group project where rewards are shared as 1/N, aligning with the payoff formula of this project, R_i/R_total, while effort is individually contributed.
Kandel & Lazear (1992) analyse how mutual monitoring, direct reciprocity, and anticipated retaliation can enforce cooperation within a group. The repetition suppresses grade inflation; without it, the mechanism collapses because agents need a future in which to retaliate. One of the key insights of this paper is that the mechanisms of peer pressure, such as reciprocity, are more effective in small groups. This justifies the 4-person structure here. Observability is necessary for mutual monitoring.
Kandel & Lazear (1992) is a theoretical paper; it does not test the model empirically, and this project is an experiment on the behaviour of the peer assessment mechanism. However, this literature is based on peers' ability to punish one another, which is only possible with repeated interaction. This explains why we expect the one-shot baseline to suffer from free-riding and misreporting.

Dana, Weber & Kuang (2007) use a modified version of the dictator game to show that ambiguity increases the likelihood that the dictator chooses the selfish option. In this game, the dictator is actively avoiding the information because otherwise it would force him to confront the moral weight of the choice. The treatment conditions the wiggle-room channel and reverses it. The cross-validation procedure in the second condition for losses removes potential ambiguity by creating an explicit, verifiable, and fair standard. This mechanism forces the player to face the moral weight of the choice. This forces the individual to confront their internal moral weight, effectively eliminating the plausible deniability that has been shown to be exploited in the modified dictator game proposed by Dana, Weber & Kuang (2007). The contrast will be observed by comparing the result to the first condition.

Bursztyn & Jensen’s (2015) research shows the importance of visibility and potential scrutiny in a choice by randomly making an educational choice public or private, thereby changing the student's behaviour. Bursztyn & Jensen’s (2015) result forms the basis for the comparison between the baseline and treated conditions, in which the only changes are the design feature and the verification structure. The difference from Bursztyn & Jensen’s (2015) paper is that the channel activated here is not the social image but the self-image, and the potential shame linked to the decision would stem from others' ability to observe it. The agent will have to observe it and will be capable of strategically ignoring it. Unlike Bursztyn & Jensen’s (2015), the second treated condition, the cross-validation result, and the credibility weight are strictly private, isolating moral self-image and ruling out social peer pressure as a confounding variable.

Andreoni & Bernheim’s (2009) paper demonstrates that in a dictator game, if a computer or mechanism can override or influence the decision, the dictator will tend to make more selfish decisions. This conclusion is the opposite of the project's expectation for the treated condition, but if the result tends to be more selfish, this is one possible answer. The fact that

fairness is supposedly ensured by a third-party actor makes the self-image channel less likely to be activated when a member makes a selfish choice. In the second condition, the cross-validation system's flag is kept private, so social image concerns remain constant across all conditions. Andreoni & Bernheim's (2009) research shows that the treated condition could be biased by the absence of social image, but including social image in the condition would conflict with moral self-image; therefore, this project won't use it. Andreoni & Bernheim's (2009) paper focuses on a dictator hiding from an audience to protect their social image. The project proposes a counter-hypothesis that tests whether this "hiding being the computer" effect still occurs when the audience is replaced by the dictator's self-image.

Experimental design:

The experiment designed in this project is intended to be conducted in a lab to increase control over time pressure, avoid the use of AI tools, and enable face-to-face identifiability. To replicate the condition of coursework. It requires 120 subjects, 60 per condition. Random session-level assignments to each condition to avoid cross-contamination

The task will consist of an effort game in which participants count zero in a 10x10 binary matrix for 5 minutes. This task design enables agents to avoid needing prior knowledge. It also gives us objective, verifiable results and generates an unambiguous score, denoted θ_j wich will act as the ground truth. At the end of the task, all four scores and the identity will be displayed before the agent is asked to complete the peer assessment. The group output is denoted by G, the sum of each score, mapped to a baseline points of 100.

This project will use a between-subjects design to limit the learning and cross-contamination effects. This experiment is based on two conditions, the first one being the baseline condition:

In the baseline condition, the four players will have to complete a peer assessment after finishing the task. Each subject will be asked about the performance of the three other players and will rate it on a scale of 1 to 5. The group's total reward will be redistributed based on these ratings. The crucial point of this baseline condition is that there is no penalty for how accurate other players' ratings are, which creates significant ambiguity and "moral wiggle room" to strategically deflate peers.

On the other hand, the treatment condition has to rate peers' performance in the same way; the difference lies in the formula that redistributes the group's total points among members. In this Treated condition, we use a mechanism that assesses the accuracy of each peer's rating based on the consensus of other players. The cross-validation is denoted by:

$w_{ij} = \max(0, 1 - \alpha|r_{ij} - C_{ij}|)$

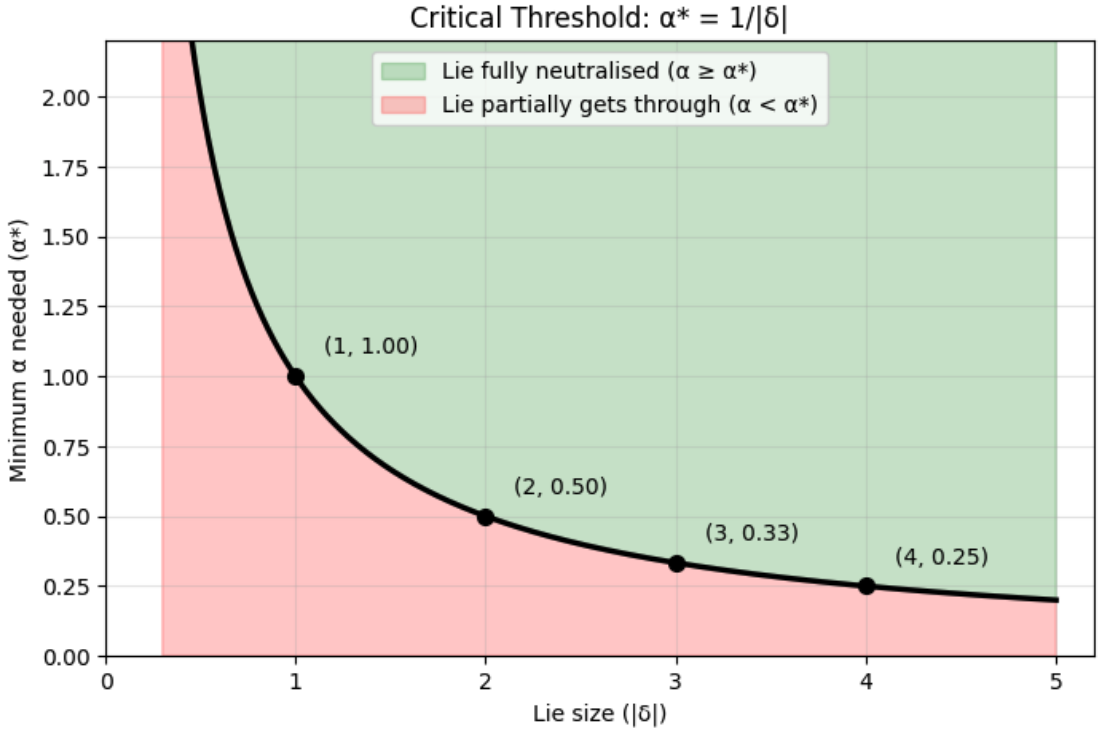
Alpha, the penalty parameter, is expected to be calibrated in a separate pilot. We're under the impression that it won't penalise honest disagreement and that it will be disclosed in the instructions. The behavioural channel activated here is moral self-image, through reduced wiggle room Dana, Weber & Kuang, 2007). In this context, honest reporting becomes a weak Nash equilibrium under unilateral deviation.

To ensure the incentive is compatible, note that all decisions are paid out in real money, so the participant takes all decisions seriously. The Peer assessment Factor (standard) determines the payment of the baseline condition with the following mechanism:

Under the PAF, their grade becomes:

This change makes the grade individual, and since it is associated with the effort exerted by student j, increased effort results in higher peer assessment scores. The student's grade is. This design is budget-balanced, meaning that no points are created; they are distributed individually based on the agent's rating.

On the other hand, the treated condition payoff is defined by a modified version of the peer assessment factor that weights the credibility of each rating. This mechanism uses a variable alpha that calibrates how strict it is. The following graphic shows the evolution of the strictness of the mechanism relative to the calibration of alpha to adapt to the different sizes of deviation:



Note that the capacity to calibrate this system is a great strength, but it requires a pilot experiment to calibrate the alpha variable by observing which types of deviation are considered missreporting.

To ensure that this experiment isolates a single causal mechanism, the researcher will randomly assign participants to either the control or treatment group. All the measures this experiment employs, such as anonymity and the one-shot nature of the game, are constant across all groups to ensure that potential behaviours cannot be attributed to task difficulty, repeated interaction, or information differences. Therby all potential differences in behaviour are attributed to the reduction of moral wiggle room.

To understand the impact and differences in behaviour between the baseline and treatment conditions, this project will observe key variables. The first key variable is the strategic deflation rate, defined as the difference between an agent's rating of a peer and that peer’s actual objective-matrix count. The hypothesis of this project is that there will be a high strategic deflation rate in the baseline condition group, and that this deflation variable will be significantly lower in the treatment group, where peer assessment is combined with a cross-

validation mechanism. The second key variable to observe is the overall group payoff efficiency, i.e., the frequency of perfectly accurate ratings.

Some potential bias might arise during the experiment. The first is the experiment demand effect. The subject might guess that the experiment is assessing people's fairness and might rate people fairly to please the experimenter. To avoid this, this project will use only neutral language in the instructions and maintain a strict no-deception policy. The second potential bias is cognitive confusion. The penalty mechanism and the calibration of alpha are complex, and the subject might make random mistakes rather than strategic choices. To address this, the consequences of the penalty mechanism will be explained in the instructions in a very simple way, with an example.

Expected result:
This project predicts higher strategic deflation in the baseline condition and lower strategic deflation in the treatment condition. Since the baseline condition uses only a standard peer-assessment factor, participants can hide behind subjectivity and exploit the “moral wiggle room” to systematically deflate their peers' ratings, which is in their own interest.
The introduction of cross-validation for peer assessment in the treatment condition will significantly reduce the strategic deflation rate. The isolation of this effect will enable this project to establish a measure that describes the pure effect of the “moral wiggle room” in the context of a group project. This prediction is based on the potential effect of the penalty mechanism, assuming the alpha variable is correctly calibrated in the pilot. If the alpha parameter is correctly calibrated, the agents will be forced to confront the moral weight of lying about their peers' effort.

Contribution:
The standard literature on collaboration has largely studied the effects of repetition, mutual monitoring, and anticipated retaliation.
This project is based on this literature and seeks to simulate the benefits of repeated collaboration by using a mix of two existing mechanisms. The results of this experiment are highly relevant because they closely reflect the conditions encountered in group projects at university. The results of such an experiment would provide direct evidence on whether an objective, transparent standard can effectively counter the threat of peer retaliation, generating a form of accountability.

Conclusion and limitation:
To conclude, this experiment uses a model that has certain limitations that are important to acknowledge and explain to improve future experimental design. The first limitation stems from the face-to-face design of this experiment. The fact that the experiment takes place face to face might partially drive decisions based on social image or fear rather than on pure self-image.
The second limitation is the reference point used in the peer assessment factor model, which is the consensus. By using the consensus as a reference point, the model does not require external monitoring, but the potential bias in some of the rankings will be taken into account in the calculation of certain consensus values. A highly organised group could collude if the collusion includes more than half of the group, because they would then constitute the consensus.
This proposal acknowledges those limitations and suggests no solution, because they stem from design choices that make this cross-validated peer assessment very well suited to the university context. Moreover, the limitations that are stated are also present in the classic

PAF, and this classic version is already used in certain universities, so we could argue that those limitations are known by universities.
The design that is offered in this proposal is applicable at the university, easily scalable, and reinforces the resistance of a mechanism that has already been adopted by several universities, offering a practical way to enforce fairness in a group project.